PROBLEM STATEMENT

GreenBuild AI

Intelligent Sustainable Construction Planner

Hackathon	HackS'US Edition V
Theme	AI in Workflow
Track	Mechanical & Civil Engineering
Duration	42 Hours	March 6 (Evening) – March 8 (Morning), 2025
Team Size	Up to 4 members
Industry Partner	CDISC Technologies

01 | Background & Industry Context

The global construction industry is responsible for nearly **40% of total carbon emissions** and consumes over 50% of all raw materials extracted worldwide. Despite the availability of proven sustainable technologies, adoption on the ground remains critically low — not due to a lack of solutions, but due to a **fundamental gap in workflow intelligence.**

Today, when an architect, structural engineer, or developer finalizes a design plan, there is no intelligent system that can analyze that plan component by component and automatically recommend the most suitable, technically validated sustainable alternative for each structural element — tailored to the building type, site conditions, climate zone, and budget.

CDISC Technologies has developed and validated a portfolio of next-generation sustainable construction solutions — including PEN Foundations, LSG Frames, and PCB Walls — that are technically proven but remain underutilized because design professionals lack a streamlined, intelligent tool to discover, compare, and integrate them into their existing workflows.

02 | The Problem

When a construction professional submits a design plan today, there is no intelligent system that can:

- Automatically identify which structural components can be replaced with sustainable alternatives
- Factor in load requirements, soil data, climate zone, material availability, and cost constraints simultaneously
- Recommend technically validated solutions backed by real engineering data
- Generate a simplified, actionable report that a client or builder can understand — without a civil engineering degree

The result: sustainable construction remains the exception, not the default. Builders revert to conventional methods simply because the decision-making process is too complex, too manual, and too fragmented.

03 | The Challenge

Design an AI-powered construction workflow tool that accepts an existing building design as input and intelligently recommends component-level sustainable alternatives — such as CDISC's PEN Foundations, LSG Frames, PCB Walls, and other validated eco-technologies — generating a simplified, measurable, and immediately actionable output for builders, architects, and clients.

04 | CDISC Technology Stack — The AI Knowledge Base

The following are CDISC-validated and industry-researched sustainable technologies that form the core recommendation database for the AI tool. Participants must map at least three of these technologies into their solution.

🏗️ PEN Foundation — Pre-Engineered Nanocomposite Foundation

What it is: A factory-engineered foundation system where structural components are prefabricated off-site using nanocomposite concrete (enhanced with nano-silica or nano-alumina additives), then assembled on-site. Eliminates on-site formwork, curing delays, and concrete waste.

How it works: Soil testing and load analysis are performed digitally. Foundation components (footings, grade beams, piers) are designed to exact specifications. Nanocomposite additives increase compressive strength by 25–40%, allowing smaller, lighter foundation profiles.

Components are bolt-connected on-site in days — not the weeks required for conventional cast-in-place foundations.

AI Applicability: Given soil type, structural loads, and site seismicity, AI can recommend PEN foundation profiles, calculate material quantities, and compare cost and carbon footprint against conventional RCC foundations in real time.

- **Carbon saving:** 30–40% reduction vs conventional RCC foundation
- **Speed benefit:** 3–5× faster installation
- **Best for:** All structure types, particularly load-sensitive and urban sites

🔩 LSG Frames — Light Steel Gauge Framing

What it is: A structural framing system using thin, cold-rolled galvanized steel sections (typically under 6mm thick) to form walls, floors, and roof structures. Sections are factory-cut using BIM/CAD-generated specifications and arrive on-site as an assembly kit.

How it works: Steel coils are cold-rolled into C-sections, Z-sections, and U-channels with pre-punched openings for wiring and plumbing. On-site assembly is purely mechanical — bolts and screws, no welding, no wet trades. Wall cavities are infilled with insulation and finished with cladding. The system is fully demountable and relocatable.

AI Applicability: AI can auto-generate LSG frame specifications from a floor plan, calculate the exact steel tonnage, optimize section sizes for structural efficiency, and compare the total embodied carbon against timber or masonry alternatives.

- **Weight:** ~1/3 the weight of timber equivalent; dramatically reduces foundation load
- **Waste:** 100% recyclable; only 2% material wastage vs 20% for timber
- **Speed:** Installation up to 50% faster than conventional methods
- **Carbon saving:** 40–60% reduction vs conventional masonry or timber framing
- **Best for:** Residential, commercial, and modular construction

🧩 PCB Walls — Precast Concrete Block / Panel Walls

What it is: Wall systems manufactured in a factory under controlled conditions — with embedded insulation (EPS or XPS core) sandwiched between concrete layers — then crane-lifted and bolted into position on-site. No mortar, no brick-laying, no waiting for walls to cure in the field.

How it works: Concrete is poured into precision molds with reinforcement mesh and insulation panels already placed. Panels are steam-cured to full strength in the factory. On-site, they are crane-lifted and mechanically connected to the structural frame. Wall finishes can be factory-applied, eliminating plastering on site.

AI Applicability: AI can recommend PCB panel types and thicknesses based on thermal zone requirements, structural loads, and fire ratings — and generate a panel schedule directly from the architectural plan.

- **Thermal performance:** Insulated sandwich panels reduce HVAC load by 30–40%
- **Construction waste:** Near-zero on-site waste; factory controlled environment
- **Carbon saving:** 25–35% reduction vs conventional brick masonry
- **Speed:** Walls erected in hours; 4–6× faster than conventional brickwork
- **Best for:** Commercial buildings, warehouses, residential, fast-track projects

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05 | Additional Validated Sustainable Alternatives

Beyond CDISC's core portfolio, the AI tool must be capable of recommending the following industry-validated technologies where appropriate:

🌍 CSEB Blocks — Compressed Stabilized Earth Blocks

Blocks made directly from on-site subsoil, mixed with 5–8% cement or lime as stabilizer, and compressed under hydraulic pressure. No kiln firing, no long-distance transport. A 50cm earth wall provides a 12-hour thermal lag — absorbing heat by day and releasing it at night for passive temperature regulation. **10–12× lower embodied energy than fired bricks; 15–40% cheaper.**

- **Best for:** Low-rise residential, rural construction, low-humidity climates
- **Limitation:** Not suitable for high-rise or high-moisture zones without waterproofing

⚡ AAC Blocks — Autoclaved Aerated Concrete

Made from fly ash (industrial waste), cement, lime, and aluminium powder. The chemical reaction during autoclaving creates millions of micro air pockets — resulting in a block that is 3× lighter than conventional bricks with exceptional thermal and acoustic insulation (up to 70% air content). Fly ash diversion from landfills is a key environmental benefit.

- **Carbon saving:** 40–50% vs clay bricks; uses recycled fly ash as primary input
- **Cost:** 10–20% cheaper than conventional bricks
- **Best for:** All building types; ideal where reduced structural load is critical

🌿 Fly Ash Bricks

A direct substitute for conventional clay bricks — manufactured from fly ash (coal combustion byproduct), cement, sand, and lime. No clay extraction, no kiln firing. 20–25% cheaper and 40–50% lower embodied carbon than red bricks, with equal or superior compressive strength.

- **Best for:** Any masonry application as a 1:1 replacement for clay bricks

🌾 Hempcrete

A bio-composite of hemp plant stalks (hemp hurd) mixed with lime and water. Hempcrete is one of the only construction materials that is **carbon negative** — sequestering approximately 110kg of CO_2 per cubic meter during the curing process. Excellent natural humidity regulation and thermal insulation (R-value of 2.4–4.8 per inch).

- **Limitation:** Non-structural — must be paired with LSG or timber frame; ideal as infill wall material
- **Best for:** Eco-conscious residential projects; passive house construction

🔩 Ferrocement Panels

Thin-shell panels (25–50mm) made from multiple layers of wire mesh plastered with high-strength cement mortar. Uses 30–50% less cement than conventional RCC. Extremely lightweight, crack-resistant, and suitable for curved or complex forms impossible with conventional brick.

- **Carbon saving:** 30–50% vs RCC; lightweight reduces overall structural requirements
- **Cost:** 30–50% cheaper than conventional construction methods
- **Best for:** Roofing, water tanks, low-cost housing, disaster-resilient shelters

06 | Technology Comparison Matrix

Technology	Best Used For	Carbon Saving	Cost vs Conventional	Construction Speed
PEN Foundation	All structures; load-sensitive sites	30–40%	Similar / slight premium	3–5× faster
LSG Frames	Residential, commercial, modular	40–60%	Slightly higher upfront	2–3× faster
PCB Walls	Commercial, fast-track projects	25–35%	Similar to conventional	4–6× faster
CSEB Blocks	Low-rise, rural, low-humidity zones	70–85%	15–40% cheaper	Similar
AAC Blocks	All building types	40–50%	10–20% cheaper	2× faster
Fly Ash Bricks	Any masonry application	40–50%	20–25% cheaper	Similar to brick
Hempcrete	Eco-residential; non-structural infill	Carbon negative	Similar / premium	Similar to masonry
Ferrocement	Roofing, tanks, low-cost housing	30–50%	30–50% cheaper	2× faster

07 | What Participants Must Build

Teams must design and demonstrate an end-to-end AI-powered construction workflow prototype that accomplishes the following five functions:

1. DESIGN INPUT PROCESSING — Accept an existing building design as input (floor plan sketch, project brief form, PDF, or structured data entry). Parse and identify the key structural components: foundation type, wall system, framing, roofing, and floors.

2. AI COMPONENT ANALYSIS — Use AI/ML models to analyse each component against the user's inputs: building type, number of floors, geographic location, soil type, climate zone, seismic zone, and budget range.

3. SUSTAINABLE RECOMMENDATION ENGINE — Map each identified component to the most suitable sustainable alternative from the CDISC technology stack and validated alternatives database. Provide ranked recommendations with justification.

4. IMPACT SCORING — For each recommendation, compute and display: estimated carbon reduction (%), approximate cost delta, construction time savings, structural compliance notes, and a composite Sustainability Score.

5. SIMPLIFIED OUTPUT REPORT — Generate a clean, non-technical summary understandable by a client, builder, or developer — explaining what changes, why they are recommended, and what the user stands to gain in cost, time, and environmental impact.

08 | Judging Criteria

Criteria	Weight
AI functionality & accuracy of sustainable recommendations	**30%**
Depth and relevance of CDISC / sustainable technology integration	**25%**
Usability and clarity of output report for non-technical users	**20%**
Scalability, real-world applicability, and engineering soundness	**15%**
Prototype quality, presentation, and demo completeness	**10%**

09 | Constraints & Ground Rules

- All sustainable alternatives recommended must be technically validated and real — no theoretical, fictional, or unproven materials
- Teams must integrate a minimum of 3 technologies from the CDISC / approved sustainable alternatives database
- The AI recommendation engine must be demonstrable with at least one complete sample input → output flow during judging
- The final output must be understandable by a person with no civil engineering background
- Teams may use any AI/ML framework, API, or technology stack
- Open-source datasets, publicly available material databases, and APIs are permitted

10 | Expected Impact if Solved

Environmental	Economic
• 30–60% reduction in per-project carbon footprint • Diversion of industrial waste (fly ash) from landfills • Passive cooling via thermal mass → reduced HVAC energy load	• 10–40% material cost reduction depending on technology mix • 2–6× faster construction → lower labour and overhead costs • Reduced maintenance costs from factory-precision components
Industry	**For CDISC Technologies**
• Democratises access to sustainable construction expertise • Bridges the gap between innovation and on-site adoption • Scalable from individual homes to large infrastructure projects	• Direct showcase of CDISC solutions to industry audience • Working prototype CDISC can develop into a commercial tool • Talent pipeline: identify and recruit top engineering innovators

Build the tool that makes sustainable construction the default, not the exception.

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